Chemistry Topic 5 — Carbon, Organic Compounds, Polymers and Fibres

Sources baked in: NCERT Class 10 (Carbon and its Compounds), Class 12 (Polymers; Chemistry in Everyday Life), Lucent General Science (Chemistry), Physics Wallah (PW) State PCS Foundation, Ghatna Chakra General Science (1990–2025 PYQs). **Weight:** ★★★ High — Recurring questions on Rayon as regenerated cellulose (not starch), Neoprene as an elastomer (not thermoplastic), Kevlar-Fullerene-Graphene discovery chronology, Teflon non-stick coatings, Bakelite electrical switches, Isoprene in natural rubber, and Sulphur vulcanization. **Last verified:** September 2026

Current Affairs (2024–2026)

Year	Event	Core Concept & Must-Score Significance
2025–2026	**Biodegradable Plastics & Biopolymers**	Mandatory phase-out of single-use petroleum-based plastics in India under Plastic Waste Management Rules. Accelerated adoption of **PLA (Polylactic Acid)** derived from fermented corn starch and **PHBV (Poly-β-hydroxybutyrate-co-β-hydroxyvalerate)**, which degrade into natural bio-assimilable CO_2 and H_2O via bacterial action within 180 days.
2024	**UPPCS Prelims 2024**	Direct question on **Rayon Origin**: Keyed that Rayon is a **regenerated cellulose fibre** obtained by chemically treating natural wood pulp or cotton linters. The option "modified starch" is an established distractor.
2022	**UPPCS Prelims 2022**	Polymer Class Matching: Identified **Neoprene** strictly as a synthetic **elastomer** (synthetic rubber), not a thermoplastic. Confirmed **Fibroin** is the natural structural protein of silk, not a synthetic polymer.
2020	**UPPCS Prelims 2020**	Tested the **Chronological Discovery Order of Carbon & Synthetic Materials**: **Kevlar (1965) → Fullerenes (C_{60}, 1985) → Carbon Nanotubes (1991) → Graphene (2004) → Black Gold (2019)**.
2019	**UPPCS Prelims 2019**	Highlighted **Kevlar** as a high-strength aromatic polyamide (aramid fibre) utilized in bulletproof vests and lightweight armor due to its high tensile strength-to-weight ratio ($5\times$ steel).

Detailed Theory Notes

5.1 Carbon Bonding, Hybridization & Hydrocarbon Families

1. Catenation and Carbon's Unique Position

Carbon occupies Group 14 and Period 2 of the Periodic Table. Its electronic configuration is $1s^2 2s^2 2p^2$. - **Catenation**: The unique property of elements to form covalent bonds with atoms of the same element, generating extended chains, branched networks, and rings.

$$\text{Catenation Capability: } C \gg Si > Ge \approx Sn > Pb$$

The exceptional catenation of carbon stems from its small atomic radius ($77\ \text{pm}$), which allows effective nuclear overlap and generates a high bond energy ($348\ \text{kJ/mol}$). Silicon forms $Si-Si$ bonds, but their lower energy ($222\ \text{kJ/mol}$) and reactivity toward oxygen make long silicon chains unstable under ambient conditions.

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      H   H
      |   |
  H - C - C - H   (Ethane: sp³, single bond, σ only, bond angle 109.5°)
      |   |
      H   H

      H   H
       \ /
        C = C     (Ethene: sp², double bond, 1σ + 1π, bond angle 120°)
       / \
      H   H

  H - C ≡ C - H   (Ethyne: sp, triple bond, 1σ + 2π, bond angle 180°)
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2. Master Hydrocarbon Taxonomy

1. **Alkanes (C_nH_{2n+2})**: - Saturated aliphatic hydrocarbons. Every carbon is sp^3 hybridized with single σ-bonds. - Chemically unreactive at room temperature; undergo free-radical substitution reactions under UV light (e.g., chlorination: $CH_4 + Cl_2 \xrightarrow{h\nu} CH_3Cl + HCl$). - Methane ($CH_4$) accounts for $> 85\%$ of natural gas and is generated by methanogenic archaebacteria in oxygen-deprived environments (paddy fields, wetlands, cattle rumens).
2. **Alkenes (C_nH_{2n})**: - Unsaturated hydrocarbons containing a $C=C$ double bond ($1\sigma + 1\pi$). - The π-bond is formed by sideways lateral overlap of unhybridized p-orbitals and is susceptible to electrophilic addition reactions (hydrogenation, halogenation). - Ethene (C_2H_4) is an endogenous plant hormone promoting fruit ripening and the monomer for polyethylene.
3. **Alkynes (C_nH_{2n-2})**: - Unsaturated hydrocarbons with a $C \equiv C$ triple bond ($1\sigma + 2\pi$). - Acetylene (C_2H_2) burns in pure oxygen to yield an intensely hot flame ($\approx 3200^\circ C$), utilized in oxy-acetylene welding. Terminal alkynes have weakly acidic acetylenic protons that react with heavy metal salts ($AgNO_3, Cu_2Cl_2$) to form insoluble explosive acetylides.
4. **Aromatic Hydrocarbons (Arenes)**: - Planar cyclic conjugated compounds containing $(4n+2)$ delocalized π electrons (Hückel's Rule). - Benzene (C_6H_6) exhibits extraordinary thermodynamic stability due to resonance energy ($\approx 150\ kJ/mol$). It undergoes electrophilic aromatic substitution rather than addition reactions, preserving the intact aromatic π-system.

5.2 Allotropes of Carbon & Advanced Nanostructures

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                                ┌── Diamond: 3D sp³, tetrahedral, hardest natural substance, insulator.
                                ├── Graphite: 2D sp², hexagonal sheets, electrical conductor, solid lubri
Carbon Allotropes (Pure Carbon) ─┼── Fullerenes (C₆₀): 0D cage, 20 hexagons + 12 pentagons ("buckyball").
                                ├── Carbon Nanotubes (CNTs): 1D tubules, high tensile strength, ballistic
                                └── Graphene: 2D monolayer, single atom thick, honeycomb lattice, transpa
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1. Fullerenes (C_{60})

- Discovered in 1985 by laser vaporization of graphite by Kroto, Curl, and Smalley.
- Consists of 60 equivalent carbon atoms arranged in a spherical cage resembling a soccer ball.
- Geometry: Truncated icosahedron containing **20 hexagonal rings and 12 pentagonal rings**.
- *Isolated Pentagon Rule (IPR)*: No two pentagons share a common edge; every pentagon is entirely surrounded by five hexagons to minimize steric ring strain.
- Each carbon atom is sp^2 hybridized with slight pyramidal distortion to accommodate spherical curvature.

2. Graphene

- A single isolated two-dimensional layer of carbon atoms arranged in a planar hexagonal honeycomb lattice.
- Electrons move across the lattice as massless Dirac fermions with Fermi velocities near $10^6\ \mathrm{m/s}$, enabling ballistic electronic transport.
- Extreme thermal conductivity ($\approx 5000\ \mathrm{W \cdot m^{-1} \cdot K^{-1}}$), Young's modulus of $1\ \mathrm{TPa}$, and intrinsic tensile strength exceeding $130\ \mathrm{GPa}$ ($> 100\times$ stronger than structural steel).

5.3 Polymer Science: Reaction Mechanisms & Mechanical Classes

1. Polymerization Regimes

- **Addition Polymerization**: Monomers containing multiple bonds join without the loss of any small molecules. The empirical formula of the resulting polymer is identical to that of the starting monomer (e.g., $nCH_2=CH_2 \rightarrow [-CH_2-CH_2-]_n$).
- **Condensation Polymerization**: Intermolecular reactions between bi-functional or poly-functional monomers accompanied by the elimination of small byproduct molecules (H_2O, HCl, NH_3). The elemental composition of the polymer differs from that of the monomers (e.g., Nylon-6,6, Terylene, Bakelite).

2. Master Mechanical Classification

[Elastomers]	[Thermoplastics]	[Fibres]	[Thermosetting Plastics]
Weakest forces (van der Waals); coiled chains; elastic stretch. (Rubber, Neoprene)	Intermediate forces (dipole/dispersion); reversibly meltable. (PE, PVC, Polystyrene)	Strongest forces (H-bonds); high tensile strength, minimal stretch (Nylon, Terylene, Kevlar)	Permanently cross-linked 3D networks; non-melting; non-recyclable. (Bakelite, Melamine)

5.4 Master Plastics, Resins & Synthetic Fibres

1. Bakelite (Phenol-Formaldehyde Condensation Resin)

- Phenol reacts with formaldehyde in the presence of an acid (H^+) or alkaline (OH^-) catalyst via electrophilic aromatic substitution to form ortho- and para-hydroxymethylphenols.
- Linear condensation yields **Novolac**, a fusible soluble polymer used in protective varnishes.
- Heating Novolac with hexamethylenetetramine (source of additional formaldehyde) forms methylene bridges ($-CH_2-$) that cross-link the chains into **Bakelite**, a rigid, infusible, thermosetting three-dimensional polymer network.
- Standard industrial application: **Electrical wall switchboards, plugs, appliance sockets, and heat-resistant cookware handles**.

2. Teflon (Polytetrafluoroethylene, PTFE)

- Polymerized from tetrafluoroethylene ($CF_2=CF_2$) using ammonium persulphate catalyst at elevated pressures:

$$n(CF_2=CF_2) \rightarrow [-CF_2-CF_2-]_n$$

- The high $C-F$ bond dissociation energy ($485\ \mathrm{kJ/mol}$) and dense protective fluorine sheath render Teflon chemically inert against boiling aqua regia, strong mineral acids, and oxidizing agents up to $260^\circ\mathrm{C}$.
- Exhibits the third-lowest coefficient of friction of any known solid material, making it the standard **non-stick coating on domestic kitchenware** and self-lubricating bearings.

3. Kevlar (Para-Aramid)

- Synthesized by low-temperature polycondensation of **1,4-phenylenediamine** and **terephthaloyl chloride** in an anhydrous amide solvent:

$$nH_2N - C_6H_4 - NH_2 + nClCO - C_6H_4 - COCl \rightarrow [-HN - C_6H_4 - NH - CO - C_6H_4 - CO-]_n + 2nHCl$$

- The polymer chains form a rigid, rod-like conformation that self-assembles into liquid crystalline solutions. Upon spinning, the chains align parallel to the fiber axis, locked by dense intermolecular hydrogen bonds between amide groups and $\pi - \pi$ stacking between aromatic rings.
- Provides exceptional tensile strength-to-weight ratio ($\approx 5\times$ structural steel), high cut resistance, and dimensional stability up to $450^\circ C$; standard material for **bulletproof vests, military helmets, and aerospace composites**.

5.5 Rubber Chemistry & Natural Biopolymers

1. Natural Rubber & Sulphur Vulcanization

- Natural rubber is a natural elastomer harvested as a colloidal suspension (latex) from *Hevea brasiliensis*. Chemically, it is strictly **cis-1,4-polyisoprene**:

$$[-CH_2 - C(CH_3) = CH - CH_2-]_n$$

- In the raw state, individual polyisoprene chains are held only by weak van der Waals interactions. Under mechanical stress, the chains slide past each other, leading to permanent deformation (tacky when warm, brittle when cold).
- **Vulcanization (Charles Goodyear, 1839)**:
- Heating raw rubber with **3–5% elemental sulphur** at $140^\circ C$ in the presence of ZnO catalyst.
- Sulphur forms covalent **disulphide cross-links ($-S - S-$)** across the allylic carbon positions of adjacent chains:

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- CH₂ - C(CH₃) = CH - CH₂ -
             |
          - S - S -
             |
- CH₂ - C(CH₃) = CH - CH₂ -
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- These chemical cross-links prevent chains from slipping permanently while allowing elastic stretching and snapping back, making the rubber durable and resilient for automobile tyres.

2. Synthetic Rubbers

- **Neoprene**: Prepared by addition polymerization of **Chloroprene (2-chloro-1,3-butadiene)**:

$$nCH_2 = C(Cl) - CH = CH_2 \rightarrow [-CH_2 - C(Cl) = CH - CH_2-]_n$$

  The chlorine substituent provides polarity and chemical stability, making neoprene highly resistant to degradation by mineral oils, gasoline, and ozone.
- **Buna-S (SBR)**: Copolymer of 1,3-butadiene and styrene; standard material for automobile tread tyres.
- **Buna-N (NBR)**: Copolymer of 1,3-butadiene and acrylonitrile; used for fuel hoses and oil seals.

Bilingual Terminology -- Carbon, Organic Chemistry and Polymers

English Term	Hindi Term	Key Anchor
Organic Chemistry (कार्बनिक रसायन)	कार्बनिक रसायन	Study of carbon compounds; ~10 million known compounds
Carbon (कार्बन)	कार्बन	6th element; tetravalent; forms 4 bonds; backbone of life
Hydrocarbon (हाइड्रोकार्बन)	हाइड्रोकार्बन	Compound of only H and C; alkanes, alkenes, alkynes

English Term	Hindi Term	Key Anchor
Alkane (एल्केन)	एल्केन	CnH2n+2; saturated; single bonds; e.g., methane (CH4), ethane, propane
Alkene (एल्कीन)	एल्कीन	CnH2n; one double bond; unsaturated; e.g., ethylene (CH2=CH2)
Alkyne (एल्काइन)	एल्काइन	CnH2n-2; one triple bond; e.g., acetylene (CH = CH); used in welding
Benzene (बेंजीन)	बेंजीन	C6H6; aromatic ring; discovered by Faraday (1825); structure by Kekule (1865)
Polymer (बहुलक)	बहुलक	Large molecule made of repeating monomer units
Monomer (एकलक)	एकलक	Small repeating unit that combines to form polymer
Polymerisation (बहुलकीकरण)	बहुलकीकरण	Chemical process of joining monomers to form polymer
Plastic (प्लास्टिक)	प्लास्टिक	Synthetic polymer; thermoplastic (remeltable) or thermoset (permanent)
Rubber (रबड़)	रबड़	Natural = latex of Hevea brasiliensis; polymer of isoprene
Vulcanisation (वल्केनीकरण)	वल्केनीकरण	Heating rubber with sulphur; improves strength and elasticity
Cellulose (सेलुलोज)	सेलुलोज	Natural polymer of glucose; plant cell wall; most abundant organic compound
Nylon (नायलॉन)	नायलॉन	First synthetic fibre; polyamide; Wallace Carothers (Du Pont, 1935)
Isomers (समावयवी)	समावयवी	Same molecular formula, different structural formula; e.g., butane and isobutane

One-Liner Revision -- Carbon, Organic Chemistry and Polymers

#	Fast Fact
1	Carbon = tetravalent; can form 4 bonds; catenation (chain-forming) ability
2	Allotropes of carbon: Diamond (hardest), Graphite (soft, conductor), Fullerene (C60)
3	Diamond = 3D tetrahedral; hardest; electrical insulator
4	Graphite = layered hexagonal; soft; good electrical conductor; used as lubricant and in pencils
5	Fullerene (Buckminsterfullerene C60) = 60 carbon atoms in soccer-ball shape; discovered 1985
6	Methane (CH4) = simplest alkane; natural gas; greenhouse gas
7	Ethylene (C2H4) = simplest alkene; plant hormone (fruit ripening); makes polyethylene
8	Acetylene (C2H2) = simplest alkyne; used in oxyacetylene welding; very hot flame
9	Benzene (C6H6) = simplest aromatic compound; carcinogen; aromatic ring
10	Nylon = first synthetic fibre (1935, Du Pont); polyamide; used in parachutes, stockings
11	Polyester (PET) = from ethylene glycol + terephthalic acid; soft-drink bottles, clothing
12	Bakelite = first synthetic plastic (1907, Leo Baekeland); thermosetting; phenol + formaldehyde
13	PVC (polyvinyl chloride) = monomer = vinyl chloride; pipes, wires, credit cards
14	Polythene (polyethylene) = monomer = ethylene; plastic bags, bottles
15	Teflon (PTFE) = polytetrafluoroethylene; non-stick cookware; very low friction

#	Fast Fact
16	Natural rubber = polymer of isoprene (C5H8)n; from Hevea brasiliensis
17	Vulcanisation = rubber + sulphur at 140 degC; makes rubber stronger and elastic (Goodyear, 1839)
18	Cellulose = polymer of beta-glucose; plant cell wall; not digestible by humans
19	Starch = polymer of alpha-glucose; energy storage in plants; digested by amylase
20	Proteins = polymer of amino acids; peptide bonds; 20 standard amino acids
21	DNA = polymer of nucleotides (deoxyribose + phosphate + base)
22	Soap = sodium/potassium salt of fatty acids; amphipathic molecule (hydrophilic + hydrophobic end)
23	Biodegradable plastics = break down by microorganisms; e.g., PLA (polylactic acid)
24	Micro-plastics = plastic particles < 5 mm; accumulate in food chain; major environmental threat
25	Rayon = regenerated cellulose fibre; "artificial silk"; from wood pulp
26	LPG = mainly propane (C3H8) + butane (C4H10); ethyl mercaptan added as odorant
27	CNG = compressed natural gas (mainly methane CH4); cleaner than diesel
28	Ethanol (C2H5OH) = ethyl alcohol; from fermentation; used as fuel additive (E10 = 10% ethanol)
29	Methanol (CH3OH) = wood alcohol; poisonous; causes blindness and death
30	Chloroform (CHCl3) = trichloromethane; used as anaesthetic historically; now solvent